11 SYSTEMS OF EQUATIONS AND INEQUALITIES

Consider an Example 🚕

"A ride app charges Rs.200 base fare + Rs.20 per km."

Cost=200+20x
What stays fixed? (base fare)
What changes? (distance → **variable**)
How does cost increase? (20 per km)

Now translate:

Cost=200+20x
y = 200+20x

Slope-Intercept Form

The equation

$$y = mx+b$$

slope

$$y = 20x + 200$$

intercept

is called the **slope-intercept form** of an equation of a line.

The letter *m* represents the slope and

b represents the *y*-intercept.

Example: Ride fare

"A ride costs Rs.200 base fare + Rs.20 per km"

$$y=20x+200$$

- What is **Rs 200**?

☞ "Cost when you don't travel at all"

➞ This is the **intercept**

- What is **Rs 20**?

☞ "Extra cost for each km"

➞ This is the **slope**

Positive slope 📈

Money increasing

Distance increasing

Example:

$$y = 200 + 20x$$

Negative slope 📉

Battery decreasing

Temperature dropping

Example:

$$y = -5x + 30$$

◆ Slope (m)

☞ “How fast something is increasing or decreasing”

Rs.20 per km

Marks increasing per hour of study

Distance covered per hour (speed!)

◆ Intercept (b)

☞ “Starting value when x = 0”

Starting money

Entry fee

Initial temperature

In the above case

$y=200 + 20x$

Put $x=o$

$y= 200 + 20 (0)$

$y= 200$

Lets Revise

“What happens when x = 0?” → intercept

“How much does y change when x increases?” → slope

$y = 20x + 200$

For different values of x (distance) you can get different values of y (cost)

Distance (Km)	Cost (Rs)
0	200
1	220
2	240

Example 1:

The cost of leasing a car is $2,311 for a down payment and processing fee plus $276 per month

Given:

Down payment = $2311 (fixed)

Monthly cost = $276 /month (variable)

Let:

x = number of months

Form equation:

$$y = 276x + 2311$$

Example 2:

You have just bought a new Sony 55” 3D television set for $1,600. The value of the television set decreases by $250 per year

Given:

Original value = $1600

Decrease per year = $250

Let:

x = number of years

Form equation:

$$y = -250x + 1600$$

LINEAR EQUATION

This is a linear equation. It is a linear equation because there are no terms involving x^2 , y^2 or any higher powers of x and y. The only terms we have got are terms in x, terms in y and some numbers. So this is a linear equation or we can say /slope – intercept form of equation

i.e. $y= mx + b$

$$y = 2x - 3$$

GENERATE THE VALUES

Suppose we choose a value for x, say $x = 1$, $y = 2 \times 1 - 3 = -1$

Suppose we choose a different value for x, say $x = 2$, $y = 2 \times 2 - 3 = 1$

Suppose we choose another value for x, say $x = 0$, $y = 2 \times 0 - 3 = -3$

For every value of x we can generate a value of y.

PLOTTING THE POINTS

We can plot these as points on a graph.

We can plot the first as the point $(1, -1)$.

We can plot the second one as the point $(2, 1)$,

and the third one as the point $(0, -3)$ and so on.

Plotting the points on a graph, as shown in Figure 1, we see that these three points lie on a straight line.

This is the line with equation $y = 2x - 3$. It is a straight line and this is another reason for calling the equation a linear equation.

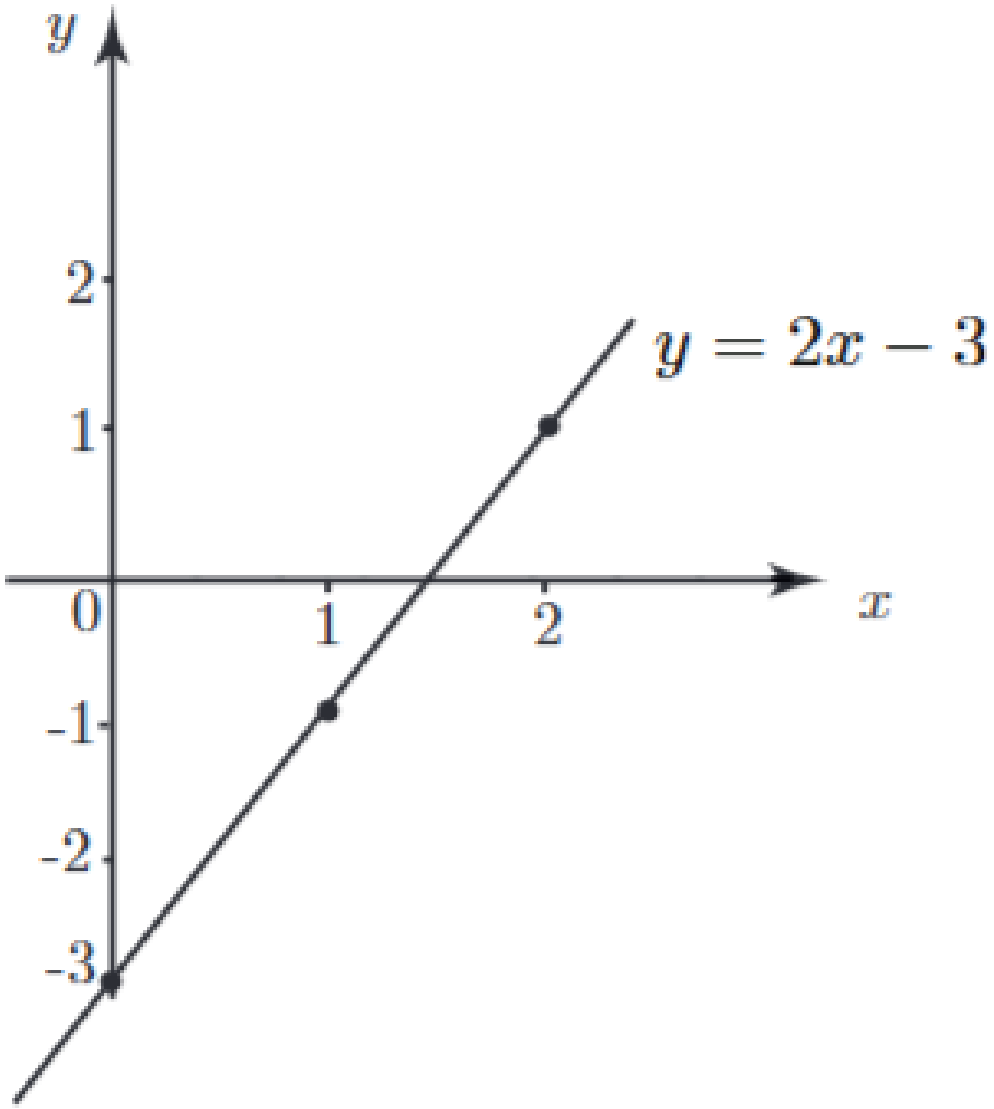


Figure 1. Graph of $y = 2x - 3$.

YOUR TURN

Office equipment was purchased for $20,000 and will have a scrap value of $2,000 each year. If its value is depreciated linearly, find the linear equation that relates value (y) in dollars to time (x) in years, highlight the slope and intercept.

SYSTEM OF LINEAR EQUATION

"A **system of linear equations** is when we have **two situations happening at the same time**,
and we want to find a value that satisfies both."

☞ In simple words:
"Find the point where two things become equal."

🚕 Example 1: Ride Apps

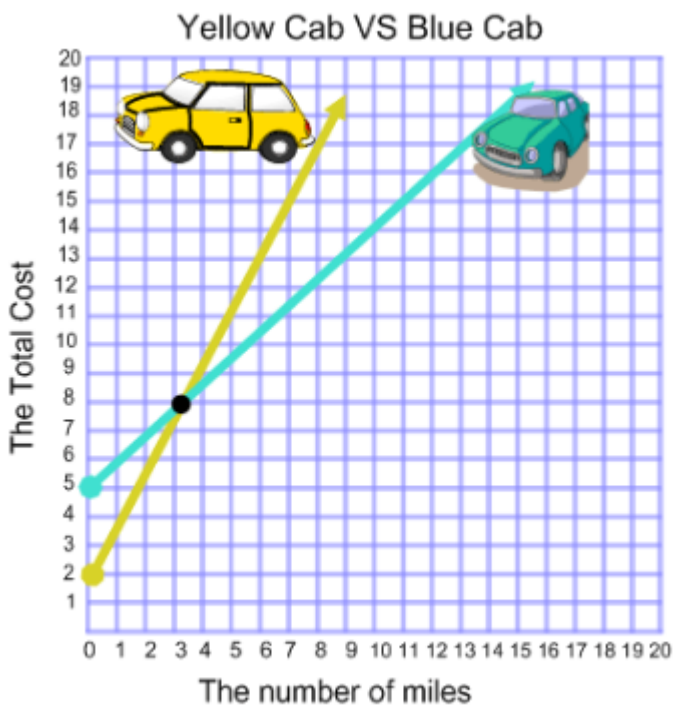
Careem: Rs 100 + Rs 20/km

Uber: Rs 200 + Rs 15/km

Can you create equation??

y= 100 +20x

y= 200 + 15x



👉 When do both cost the same?

“A system of equations helps us compare two situations and find when they become equal.”

Simple Analogy

“Imagine a lock with 3 numbers.
Each clue gives you a condition.
To open the lock, all clues must be true at the same time.”
👉 That is a **system of equations**

A system of linear equations consists of two or more linear equations made up of two or more variables such that all equations in the system are considered simultaneously. To find the unique solution to a system of linear equations, we must find a numerical value for each variable in the system that will satisfy all equations in the system at the same time.

SYSTEMS OF LINEAR EQUATIONS IN TWO VARIABLES

Consider the example $2x + y = 15$

$3x - y = 5$

The solution to a system of linear equations in two variables is any ordered pair that satisfies each equation independently. In this example, the ordered pair (4, 7).

We can verify the solution by substituting the values into each equation to see if the ordered pair satisfies both equations.

$2(4) + 7 = 15$ (TRUE)

$3(4) - 7 = 5$ (TRUE)

YOUR TURN

Determine whether the ordered pair (8,5) is a solution to the following system.

$5x - 4y = 20$

$2x + 1 = 3y$

Types of Linear Systems

There are three types of systems of linear equations in two variables.

- An independent system has exactly one solution pair solution. The point where the two lines intersect
- An inconsistent system has no solution. Notice that the two lines are parallel and will never intersect.
- A dependent system has infinitely many solutions. The lines are coincident. They are the same line, so every coordinate pair on the line is a solution to both equations.

Figure 2 compares graphical representations of each type of system.

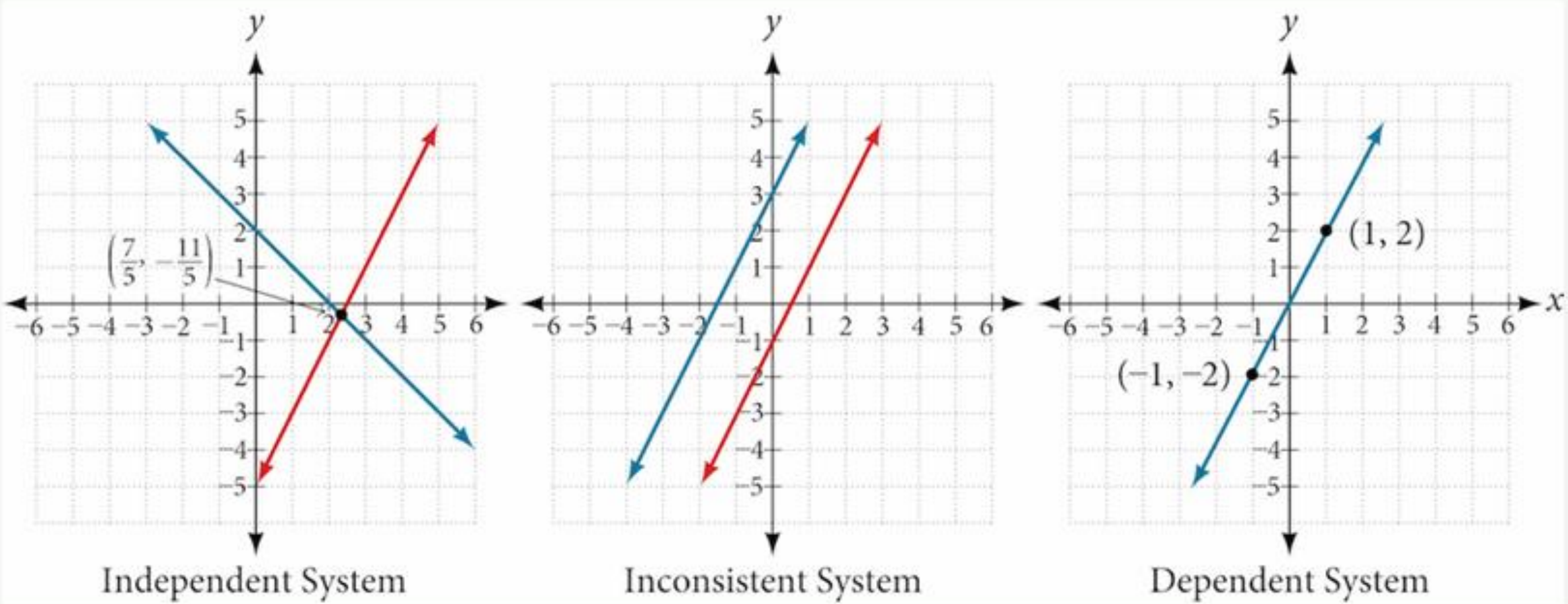

y
x
$\left(\frac{7}{5}, -\frac{11}{5}\right)$
(1, 2)
(−1, −2)
Independent System
Inconsistent System
Dependent System

Figure 2

Solving Systems of Equations in Two Variable

$$3x + 5y = -11$$

$$x - 2y = 11$$

$$x - 2y = 11$$

$$-3(x - 2y) = -3(11) \qquad \text{Multiply both sides by } -3.$$

$$-3x + 6y = -33 \qquad \text{Use the distributive property.}$$

Now, let's add them.

$$\begin{array}{r} 3x + 5y = -11 \\ -3x + 6y = -33 \\ \hline 11y = -44 \\ y = -4 \end{array}$$

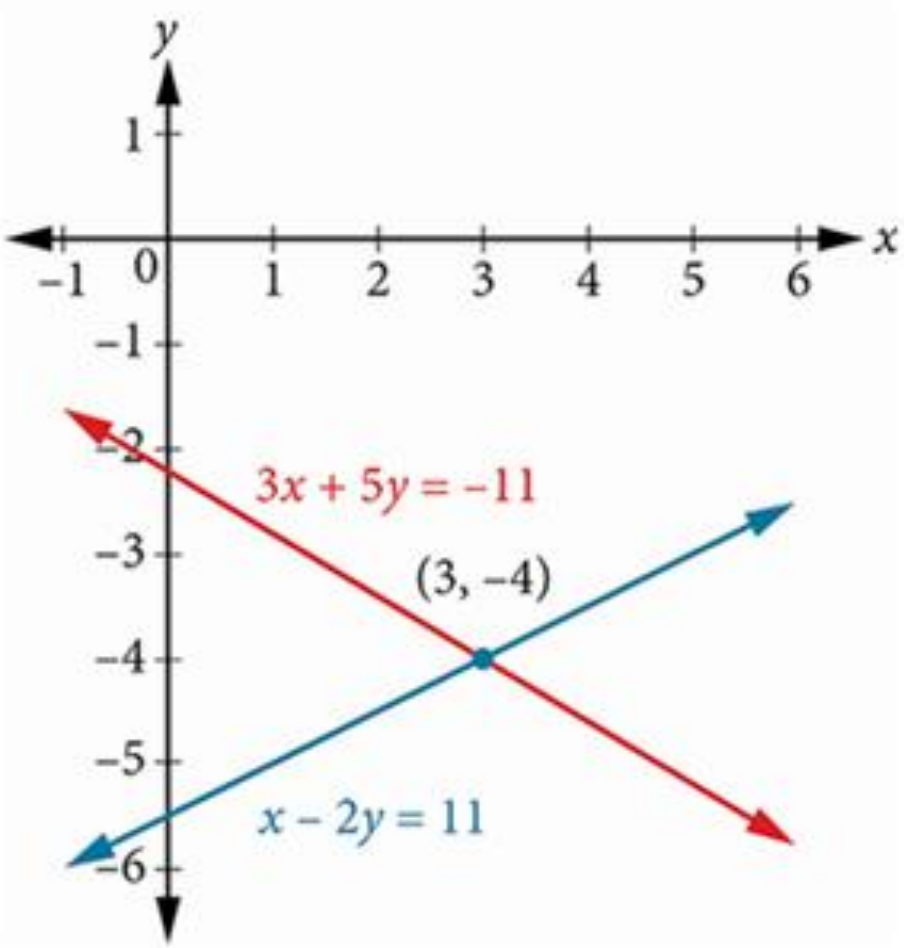
For the last step, we substitute $y = -4$ into one of the original equations and solve for x.

$$3x + 5y = -11$$
$$3x + 5(-4) = -11$$
$$3x - 20 = -11$$
$$3x = 9$$
$$x = 3$$

Our solution is the ordered pair $(3, -4)$.

Check the solution in the original second equation.

$$x - 2y = 11$$
$$(3) - 2(-4) = 3 + 8$$
$$11 = 11 \qquad \text{True}$$


y
x
1
0
−1
1
2
3
4
5
6
−1
−2
−3
−4
−5
−6
3x + 5y = −11
(3, −4)
x − 2y = 11

YOUR TURN

Solve the system of equations

$2x - 7y = 2$

$3x + y = -20$

Identifying Inconsistent Systems of Equations Containing Two Variables

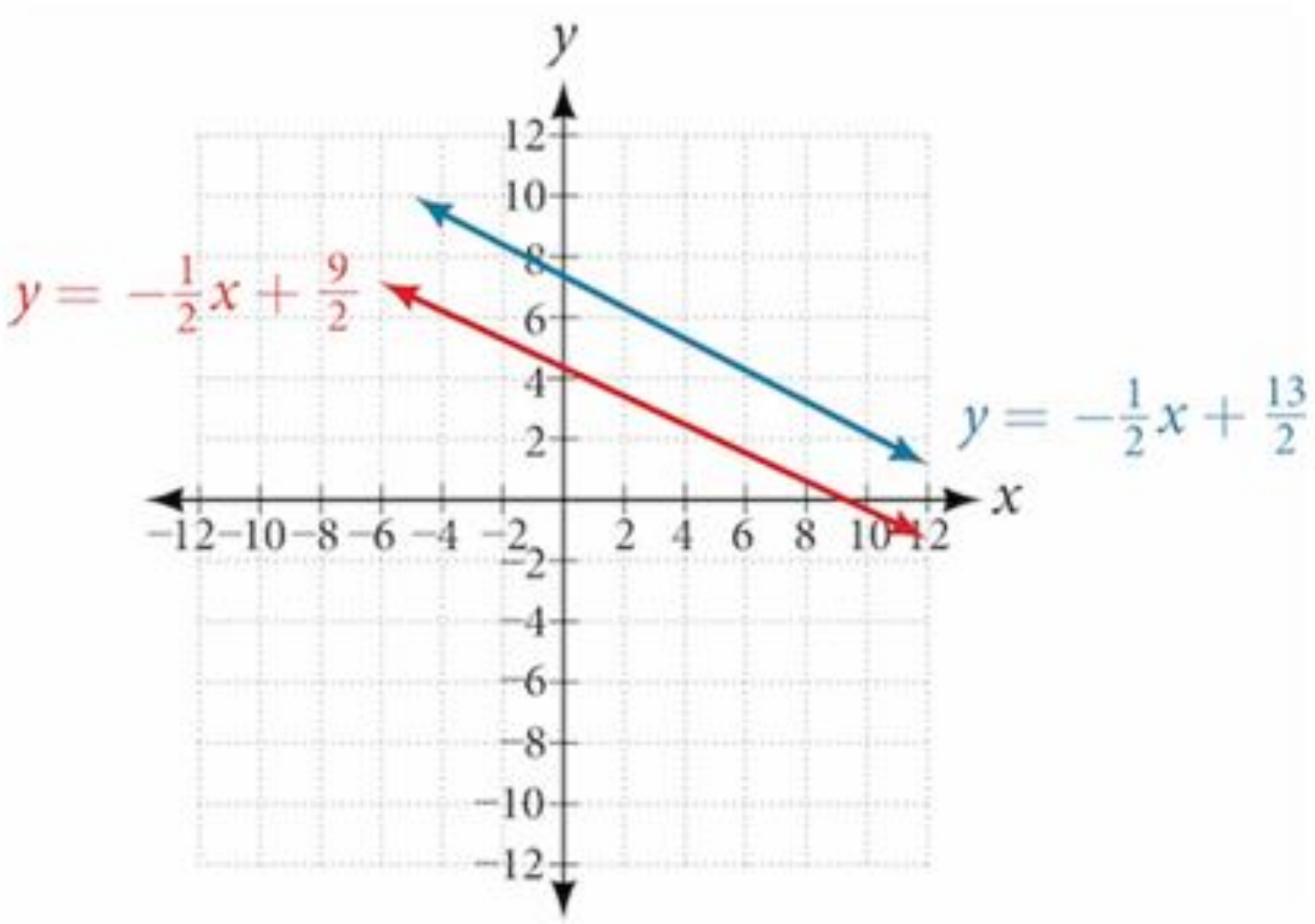
Solving an Inconsistent System of Equations

Solve the following system of equations.

$$x = 9 - 2y$$
$$x + 2y = 13$$

$$
\begin{aligned}
x + 2y &= 13 \\
(9 - 2y) + 2y &= 13 \\
9 + 0y &= 13 \\
9 &= 13
\end{aligned}
$$

Clearly, this statement is a contradiction because $9 \neq 13$. Therefore, the system has no solution.


y
x
12
10
8
6
4
2
−2
−4
−6
−8
−10
−12
−12 −10 −8 −6 −4 −2 2 4 6 8 10 12
$y = -\frac{1}{2}x + \frac{9}{2}$
$y = -\frac{1}{2}x + \frac{13}{2}$

YOUR TURN

Solve the following system of equations in two variables.

$2y - 2x = 2$

$2y - 2x = 6$

Expressing the Solution of a System of Dependent Equations Containing Two Variables

Finding a Solution to a Dependent System of Linear Equations

Find a solution to the system of equations using the addition method.

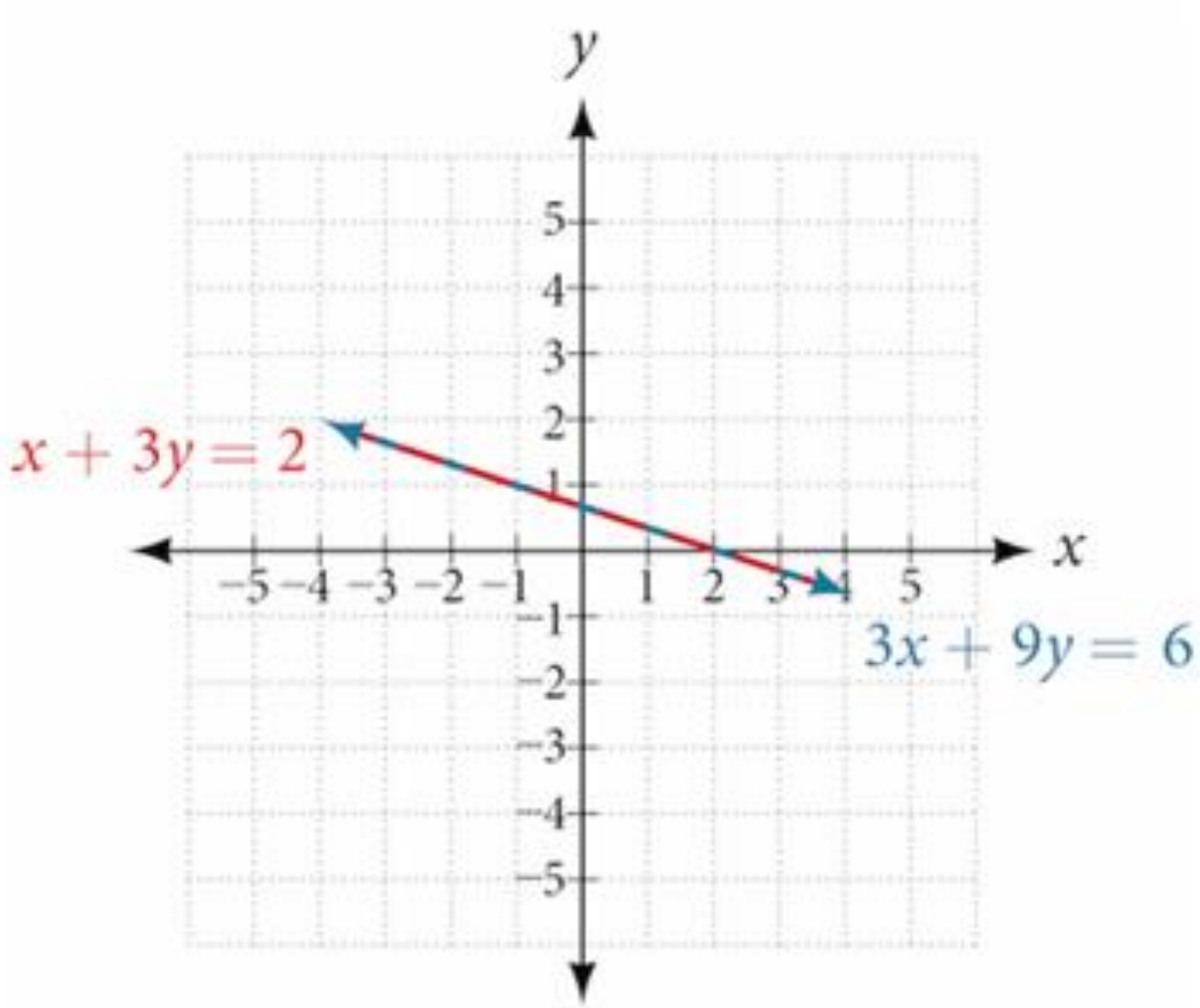
$$x + 3y = 2$$
$$3x + 9y = 6$$

$$x + 3y = 2$$
$$(-3)(x + 3y) = (-3)(2)$$
$$-3x - 9y = -6$$

Now add the equations.

$$\begin{array}{rl} -3x - 9y & = -6 \\ + \ \ 3x + 9y & = 6 \\ \hline 0 & = 0 \end{array}$$

We can see that there will be an infinite number of solutions that satisfy both equations.


y
x
5
4
3
2
1
−1
−2
−3
−4
−5
−5 −4 −3 −2 −1
1 2 3 4 5
x + 3y = 2
3x + 9y = 6

YOUR TURN

Solve the following system of equations in two variables

$y - 2x = 5$

$-3x + 6x = -15$

PRACTISE QUESTIONS

Use elimination to solve the following pairs of simultaneous equations.

a) $\begin{array}{rcrcr} 5x & + & 3y & = & 9 \\ 2x & - & 3y & = & 12 \end{array}$ b) $\begin{array}{rcrcr} 2x & - & 3y & = & 9 \\ 2x & + & y & = & 13 \end{array}$ c) $\begin{array}{rcrcr} x & + & 7y & = & 10 \\ 3x & - & 2y & = & 7 \end{array}$

Answers

a) $x = 3, y = -2$ b) $x = 6, y = 1$ c) $x = 3, y = 1$